



MUHAMMAD PANDU WIRAKUSUMA

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Jln. Raya Maluk-Jereweh, Rumah No. 94, Desa Beru, Kec. Jereweh, Kab. Sumbawa Barat, 84456

With a strong technical background in Telecommunication Engineering and Electrical Engineering and proven leadership skills, I'm ready for a management role, integrating technical expertise into effective strategies. I'm dedicated to empowering teams and achieving company goals. Excited to explore management opportunities.

Work Experiences

PT. Mukti Rejo Abadi - Jakarta Timur

Jan 2022 - Jun 2022

Product Designer

As a Product Designer in a healthcare equipment company, I'm responsible for designing product packaging and collaborating in discussions related to all aspects of the company's products.

- Assisted the founders in growing the business by developing several innovations related to market trends and designing strategies for marketing medical equipment products. Developing product marketing innovations aligned with digital trends, such as updating product images for promotion on various marketplaces.

PT. Bina Sarana Medika - Jakarta Timur

Jan 2022 - Jun 2022

Member of Creative Division

As the Member of Creative Division in a healthcare equipment company, I'm responsible for planning and creating social media designs to market the company's healthcare products.

- Assisted the branch company in growing its business by developing several innovations related to market trends and designing strategies for marketing medical equipment products, specifically for B2C sales.

PT. Integrated Sustainable Risk Solutions Indonesia (ISRSI) - Bandung City

Sep 2022 - Dec 2022

3D Designer (Intern)

As a 3D Designer (Intern) in an IT and Media company, I design IT and media-related products for the company's clients, blending creativity and technology.

- Assisted in the creation of procedural videos and 3D advertisements for multiple companies, focusing on delivering high-quality content that effectively communicates complex processes and promotes products in an engaging and visually appealing manner.

Tsuki Software & Media House - Bandung City

Oct 2021 - Oct 2025

Co Founder / UI Designer / Sales Engineer

As a Co-Founder, I orchestrate company operations by integrating strategic leadership with hands-on technical expertise. I drive the creative direction as a UI Designer to build intuitive, user-centric interfaces while concurrently serving as a Sales Engineer to bridge the gap between complex IT solutions and client requirements. By blending creative vision with technical strategy, I ensure that our media and technology offerings are both innovative and aligned with long-term business growth.

- Established a freelance community focused on IT solutions and media, where I played a key role as a negotiator and contributed to UI/UX design and other creative processes. Successfully completed projects such as website development, machine learning models for forecasting, and student yearbooks, showcasing a diverse range of expertise and a commitment to delivering innovative solutions.

Educational Experiences

Universitas Telkom - Bandung, Indonesia

Aug 2019 - Aug 2023

Bachelor of Telecommunication Engineering, 3.44/4.00

- Awarded a Fast-Track Scholarship with a minimum GPA requirement of 3.25.
- Actively participated in various organizations and laboratories.
- Engaged in several activities, including national and international seminars.

Universitas Telkom - Bandung, Indonesia

Aug 2023 - Present

Master of Electrical Engineering, 3.82/4.00

- Received a prestigious scholarship with a minimum GPA requirement of 3.75, reflecting strong academic performance and dedication.
- Actively involved in various organizations and research laboratories, contributing to collaborative projects and team initiatives.
- Participated in a wide range of activities, including attending and presenting at both national and international seminars, showcasing a commitment to personal and professional development.

Organizational Experiences

Nano Satellite Laboratory, Telkom University - Bandung

Dec 2019 - Aug 2023

Member of ADCS Sub-Division

Contributed to the ADCS division of the Nano Satellite Laboratory, focusing on satellite attitude control systems, signal processing, and system design to support nanosatellite technology development.

Smart Data Sensing (SDS) Laboratory, Telkom University - Bandung

Aug 2023 – Present

Head of Audit & HR

Conducted research and projects in the Smart Data Sensing Laboratory, specializing in data acquisition, sensor integration, and smart systems for innovative IoT solutions and intelligent data processing.

Al-Fath Campus Islamic Organization, Telkom University - Bandung

Aug 2019 - Jun 2020

Member of External Division

Played a vital role in the External Division, organizing collaborations, outreach programs, and inter-organizational activities to strengthen community engagement and uphold organizational values.

Telkom University Photography Club - Bandung

Dec 2020 - Aug 2023

Member and President Club (2021-2022)

Actively participated in the Photography Club, enhancing technical photography skills and contributing to various projects, exhibitions, and documentation of university events.

Regional SMEs of Samalowa, Telkom University - Bandung

Aug 2019 - Aug 2020

Member of Design & Documentation Division

Worked in the Design and Documentation Division, responsible for creating visual content, managing event documentation, and developing creative materials for organizational activities.

Telco Hooligans, HMTT Telkom University - Bandung

Aug 2020 - Aug 2023

Member of Creative Division

Contributed to the Creative Division of Telco Hooligans, a semi-autonomous organization under HMTT, by designing creative campaigns, producing promotional materials, and supporting event branding initiatives.

Project Result, Courses, Etc

- **Modules Taken (2021):** Englist Course, LC Language Center JBI Group: Participated in the intensive and integrated English Course at LC Language Center JBI Group, covering grammar, pronunciation, vocabulary, speaking, mind mapping, discussion, presentation, public speaking, debate, and job interview preparation. Utilized diverse learning methods such as classroom instruction, English area dormitory immersion, continuous learning strategies, enjoyable experiences, and practical applications.
- **Modules Taken (2022):** Data Analyst Program, Algoritma Data Science School - Training Center: Participated in the Data Analyst Program at Algoritma Data Science School - Training Center, focusing on essential data analytics skills. Received intensive training in SQL, R or Python, and developed strong capabilities in data visualization and modeling using Power BI and Tableau.
- **Modules Taken (2022):** UI UX Program, Alterra Academy: Participated in the UI/UX Program at Alterra Academy under Kampus Merdeka. Gained skills in user interface and user experience design through learning modules, workshops, and practical assignments. Completed a capstone project, showcasing proficiency in delivering professional UI/UX solutions.
- **Projects (2021): DigiTelco - Information Portal for Telecommunications Engineering Students:** Developed DigiTelco, an information portal for Telecommunications Engineering undergraduates, similar to Facebook. The platform provided news, updates, and resources, promoting community engagement. Served as the UI/UX Designer and Logo Designer, ensuring an intuitive interface and cohesive visual identity.

Job Desk (Tsuki Software & Media House Project)

- Logo & Visual Identity Design
 - Spearheaded the visual identity development for the DigiTelco platform, including the creation of a professional and modern logo from concept to final vector execution.
 - Translated the community's values into a cohesive visual language, ensuring the brand identity resonated with Telecommunications Engineering students.
 - Maintained design consistency across various digital touchpoints to establish a recognizable and trustworthy platform presence.
- User Research & Usability Testing
 - Conducted end-to-end Usability Testing with target users to identify navigation friction, accessibility issues, and overall user satisfaction.

- Gathered and analyzed qualitative user feedback, translating complex user pain points into actionable design improvements.
 - Iterated on UI/UX wireframes and prototypes based on testing results, successfully enhancing the platform's intuitive flow and increasing user engagement.
 - Collaborated on the information architecture to ensure news, updates, and resources were easily discoverable, mimicking a seamless social-media-like experience.
- **Projects (2022): Design Packaging:** Served as a Product Designer for a healthcare equipment company, where I specialized in the end-to-end packaging design for medical supplies, including masks, aprons, and syringes. I collaborated closely with the founders to drive business growth by developing marketing innovations and digital strategies. This included optimizing product visual assets for various e-commerce marketplaces to enhance brand presence and ensure alignment with current market trends.

Job Desk (PT. Mukti Rejo Abadi Internal Job)

- Product Packaging & Visual Design:
 - Spearheaded the end-to-end packaging design process for a diverse range of medical supplies, including surgical masks, protective aprons, and syringes.
 - Developed professional and sterile visual layouts for product boxes, ensuring that all designs complied with medical labeling standards while maintaining a cohesive brand identity.
 - Managed the transition from concept to final print-ready assets, focusing on material durability and clear information hierarchy for healthcare professionals and consumers.
- **Projects (2022): Voting- Q - Innovative Digital Voting Platform:** An innovative project enhancing the voting experience through advanced technology and streamlined processes. It ensures efficient, secure, and accessible voting with modern algorithms and user-friendly interfaces. I played a key role as the negotiator for project deals, ensuring successful collaborations and partnerships.

Job Desk (Tsuki Software & Media House Project)

- Team Leadership & Strategic Negotiation:
 - Orchestrated the end-to-end project lifecycle, leading a cross-functional team to develop a secure, efficient, and technology-driven voting platform.
 - Served as the lead negotiator for project deals, successfully securing strategic partnerships and collaborations that were pivotal to the project's implementation.
 - Managed project timelines and resource allocation, ensuring all technical milestones were met while maintaining high standards of data security and integrity.
 - UI Design & User Accessibility:
 - Spearheaded the UI design process, creating a modern and intuitive interface that simplifies complex voting procedures for a diverse range of users.
 - Prioritized digital accessibility and inclusivity, ensuring the platform's visual elements and navigation were optimized for ease of use across various devices.
 - Translated advanced algorithmic requirements into user-friendly layouts, balancing sophisticated backend security with a seamless and transparent frontend experience.
 - Developed high-fidelity prototypes and conducted design reviews to ensure the final product aligned with the project's vision of modernizing the democratic process.
- **Projects (2022): NPK Information Website:** Led the development of the NPK Information Website, playing a key role as the negotiator for project deals. Successfully facilitated collaborations and partnerships, ensuring the project's success.

Job Desk (Tsuki Software & Media House Project)

- Project Negotiation & Partnership Management:
 - Served as the primary negotiator for project deals, successfully securing strategic partnerships and collaborations essential for the project's launch.
 - Facilitated contract discussions and project scoping, ensuring that all agreements aligned with both technical capabilities and stakeholder expectations.
 - Managed stakeholder relationships, acting as the key bridge between external partners and the development team to ensure smooth project delivery and resource acquisition.
 - UI Design & Information Architecture:
 - Designed a clean and intuitive user interface for the NPK Information hub, specifically optimized for presenting complex agricultural data in an easy-to-digest format.
 - Created high-fidelity wireframes and interactive prototypes, ensuring the visual layout prioritized user accessibility and seamless navigation.
 - Developed a cohesive visual identity for the platform, ensuring that the presentation of technical information remained engaging and professional for the target audience.
 - Collaborated on the information hierarchy, organizing data-heavy content to improve the overall user experience and information discoverability.
- **Projects (2022): 3D Video Presentation:** Developed 3D-driven media products, including procedural videos and advertisements for various corporate clients. I played a key role in creating a high-profile 3D presentation video for Pertamina using Adobe After Effects, where I translated complex technical processes into engaging visual narratives. My work focused on blending creativity and technology to deliver high-quality content for both promotional and instructional purposes.

Job Desk (PT. Integrated Sustainable Risk Solutions Indonesia (ISRSI) Project)

- 3D Visualization & Motion Graphics:
 - Developed high-impact 3D assets and animations for corporate presentation videos, specifically for high-profile clients such as Pertamina.
 - Utilized Adobe After Effects to create sophisticated 3D motion graphics that effectively simplified and communicated complex industrial processes.
 - Produced professional-grade 3D advertisements, focusing on delivering visually appealing content that aligns with the client's corporate identity and marketing goals.
- Content Strategy & Procedural Design:
 - Translated intricate technical workflows into clear procedural videos, ensuring that technical information was accessible and engaging for the target audience.
 - Collaborated on the end-to-end creative process, from conceptualizing 3D models to final post-production, ensuring high-quality visual storytelling.
 - Blended creative design with technology to produce media products that serve both promotional and instructional purposes for large-scale energy and IT sectors.
- **Projects (2023): Warehouse Inventory Website (Internal):** Oversaw the creation of the Warehouse Inventory Website, playing a pivotal role as the project deal negotiator. Successfully orchestrated collaborations and partnerships, ensuring the project's success.

Job Desk (Tsuki Software & Media House Project)

- Strategic Negotiation & Procurement:
 - Served as the primary negotiator for the project's procurement phase, successfully securing necessary resources and internal departmental approvals.
 - Facilitated cross-functional collaborations between the technical development team and warehouse stakeholders to ensure the project scope aligned with internal operational needs.
 - Orchestrated strategic agreements that enabled the successful development of a private, high-security inventory system tailored for internal corporate use.
- Contractual & Administrative Management:
 - Managed the end-to-end legal documentation process, including the drafting, reviewing, and finalization of Memorandums of Understanding (MoUs) and project contracts.
 - Ensured full administrative compliance with internal corporate governance standards, managing all necessary paperwork to formalize partnerships and project milestones.
 - Acted as the key bridge between legal, finance, and technical departments to resolve contractual discrepancies and streamline the project's approval workflow.
- **Projects (2024): Perguruan Cikini Company Profile Website:** Led the development of the Company Profile Website for Perguruan Cikini, serving as the project deal negotiator and UI/UX Designer. Successfully facilitated collaborations, ensuring the project's success, while designing an intuitive user interface and cohesive visual identity.

Job Desk (Tsuki Software & Media House Project)

- Business Negotiation & Stakeholder Management:
 - Lead the negotiation and acquisition of the project deal, successfully aligning the institution's vision with technical execution requirements.
 - Facilitated high-level collaborations with stakeholders from Perguruan Cikini to define project scopes, timelines, and key deliverables.
 - Managed the contractual process and project briefing, ensuring all institutional branding guidelines and communication goals were met through clear and professional agreements.
- UI/UX Design & Brand Identity:
 - Spearheaded the end-to-end UI/UX design process, creating an intuitive and accessible interface tailored for students, parents, and stakeholders.
 - Developed a cohesive visual identity that modernizes the institution's digital presence while respecting its long-standing educational heritage.
 - Designed responsive layouts and high-fidelity prototypes, focusing on seamless navigation to ensure information about programs and admissions is easily discoverable.
 - Conceptualized the website's aesthetic, ensuring a professional and trustworthy look-and-feel that aligns with the prestigious reputation of Perguruan Cikini.
- **Projects (2025): Academic Administrative & Correspondence System for Universitas Bengkulu:** Oversaw the development of the Letter Application for Universitas Bengkulu, acting as both the project deal negotiator and the UI/UX Designer. Successfully managed collaborations to ensure project completion while crafting a user-friendly interface and cohesive visual design.

Job Desk (Tsuki Software & Media House Project)

- Strategic Negotiation & Requirement Engineering:
 - Lead the successful negotiation and acquisition of the project deal with Universitas Bengkulu, navigating complex institutional procurement and administrative requirements.

- Facilitated stakeholder workshops to gather and analyze the university's specific correspondence workflows, ensuring the digital solution addressed existing administrative bottlenecks.
 - Managed the formal contractual process, including the preparation and finalization of project agreements that aligned technical delivery with the university's regulatory standards.
- **UI/UX Design & Digital Transformation:**
 - Spearheaded the UI/UX design to digitize manual paper-based processes, creating a streamlined and secure interface for document tracking, approvals, and archiving.
 - Developed an intuitive user experience tailored for a diverse user base, including administrative staff, faculty members, and students, ensuring high adoption rates across the campus.
 - Crafted a professional and cohesive visual identity that integrates the university's branding into a modern, functional dashboard.
 - Produced high-fidelity interactive prototypes to validate complex letter-routing logic, ensuring the final design was both technically feasible and user-centric.
- **Projects (2025): Wave Prediction Application:** Led the development of a Wave Prediction Application for an international client, managing both the project leadership and the frontend execution. I utilized Framer to build an intuitive visual interface, leveraging its auto-deploy features to ensure a rapid and seamless transition from design to production. My role involved bridging the gap between complex forecasting data and user-centric design while maintaining high-level coordination with global stakeholders.
 - **Job Desk (Tsuki Software & Media House Project)**
 - Leadership & International Stakeholder Management:
 - Orchestrated the end-to-end development of a wave prediction application for an international client, serving as the Team Leader responsible for project roadmaps and technical coordination.
 - Managed cross-border communications, ensuring that technical deliverables were perfectly aligned with the client's global requirements and industry standards.
 - Oversaw the integration of data-driven features, coordinating between backend requirements and frontend execution to ensure a seamless product launch.
 - Frontend Development & Rapid Deployment (Framer):
 - Spearheaded the frontend development using Framer, creating a highly responsive and modern user interface designed for real-time data visualization.
 - Optimized the development workflow by leveraging Framer's auto-deploy capabilities, significantly reducing the time from design finalization to live production.
 - Transformed complex wave forecasting data into intuitive visual elements, providing users with a clear and interactive experience for monitoring sea conditions.
- **Projects (2025): Phinisi Boat Custom Website Project:** Collaborated with a Phinisi shipbuilding firm to provide exclusive web development services for luxury boat buyers. As Project Lead, I managed the creation of bespoke digital identities for high-end maritime assets, blending luxury branding with functional web design. This initiative focused on delivering integrated digital solutions that enhanced the professional image and commercial value of each vessel in the international market.
 - **Job Desk (Tsuki Software & Media House Project)**
 - Strategic B2B Partnership & Project Management:
 - Established a strategic collaboration with a Phinisi shipbuilding company, integrating custom web development as a premium value-added service for every vessel purchase.
 - Orchestrated the end-to-end digital onboarding process for high-end clients, ensuring that each website perfectly reflected the unique specifications and luxury status of their new vessel.
 - Served as the primary liaison between the shipyard and the yacht owners to align branding visions with functional digital requirements.
 - Luxury Branding & Custom Web Development:
 - Led the design and development of bespoke UI/UX interfaces, focusing on a premium aesthetic that elevates the digital presence of traditional Phinisi boats for a global market.
 - Developed comprehensive information architectures, including technical galleries, vessel features, and integrated booking/inquiry systems to facilitate the commercialization of the assets.
 - Maintained high visual standards and brand consistency, ensuring that each website served as a powerful marketing tool for the owners to promote their luxury maritime investments.
- **Research and Development in Education (2021): Lora Patch Antenna and Cots Based Receiver Module for Disaster Mitigation M2M Communications:** Developed for the IEEE MTT-Sat Challenge, this project enhances disaster mitigation using Lora technology and COTS components for efficient M2M communications. It provides reliable, cost-effective solutions in disaster-prone areas, ensuring robust signal reception and seamless data transmission. This effort by Telkom University aims to improve real-time monitoring and coordination during disaster response operations.
 - **Tools usage:** Solid Works 2019.
 - **Lead Researcher & Competitor Project :** Nano Satellite Laboratory, Telkom University
- **Research and Development in Education (2023): Design of a Nighttime Livestock Motion Detection System Using Depth-Sensing Digital Imaging:** Developed a motion detection system for livestock at night using depth-sensing digital imaging for my

undergraduate thesis at Telkom University. The system employs the Frame Difference algorithm, achieving 76.15% accuracy, and allows remote monitoring of livestock behavior, providing data on sleep patterns, feeding habits, and health issues.

- **Models Implemented:** Yolo V3.
- **Lead Researcher & Competitor Project:** Bachelor Degree (Thesis), Telkom University
- **Research and Development in Education (2024): Implementation of Ensemble Machine Learning with Voting Classifier for Reliable Tuberculosis Detection Using Chest X-ray Images with Imbalance Dataset:** Developed a TB detection system using chest X-ray images and Machine Learning. Employed SVM and Random Forest models with 4200 data points, integrating them with Ensemble Learning and a Voting Classifier. Achieved 99.1% accuracy, 99.3% AUC, 99% recall, 98% precision, and 98% F1-score. Successfully managed imbalanced data, demonstrating the robustness of the approach.
 - **Models Implemented:** Support Vector Machine (SVM), Random Forest, Voting Classifier.
 - **Lead Researcher & Competitor Project:** Smart Data Sensing (SDS) Laboratory, Telkom University
- **Research and Development in Education (2024): Sentiment Analysis of Comments in Digital Banking Applications (Internal Research):** Conducted research at the Smart Data Sensing (SDS) Laboratory to analyze user comments in digital banking apps for five leading banks. Utilized Naive Bayes and Logistic Regression to identify trends in user feedback, offering insights into customer satisfaction and areas for improvement.
 - **Models Implemented:** Naive Bayes and Logistic Regression
 - **Lead Researcher & Competitor Project:** Smart Data Sensing (SDS) Laboratory, Telkom University
- **Research and Development in Education (2025): Deep Learning Prediction Of Shallot Prices To Assist The Government In Policy Formulation:** Conducted advanced research focused on Deep Learning applications for predicting shallot prices to support government policy formulation. I developed predictive models to analyze market volatility, providing data-driven insights for agricultural authorities to stabilize commodity prices. This project bridged the gap between complex neural network architectures and strategic economic planning, ensuring more accurate and actionable forecasting for national food security.
 - **Models Implemented:** ARIMA, SARIMA, Prophet (Univariate), SARIMAX, LSTM (Multivariate)
 - **Lead Researcher & Competitor Project:** Master Degree (Thesis), Telkom University